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➤ Optimization algorithms for Artificial Intelligence: Introduction

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➤ Outline

- What is this class about?
- Who am I?
- What is optimization?
- What is Artificial Intelligence?
- Is optimization a kind of Artificial Intelligence?
- What is the relationship between AI and optimization?



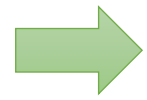
➤ What is this class about?

- Optimization!
 - Overview of optimization techniques, when/how to use them
 - How these techniques power modern Artificial Intelligence
 - Optimization to improve performance of AI methods
- At the end of the class, you should know
 - What techniques are more appropriate for different problems
 - How (several) AI systems work, especially for Machine Learning
 - Hyperparameter optimization for your AI applications
 - Keywords for further research (e.g. Neuro-symbolic, AutoML)



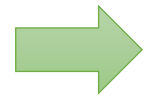
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➤ (Tentative) Schedule

April 13, sc.071

- Optimization: introduction
- Continuous optimization

Exercises

- Linear programming

April 15, sc.071

- Discrete optimization

Exercises

- Multi-objective optimization

Exercises

- Optimization of structures

Exercises

April 17, sc.071

- Optimization in ML

Exercises

- Hyperparameter optimization

April 17, sc.071

Discussion

- Recent developments

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Prepare a few slides to **present your problem**, then we discuss how to frame it as an **optimization problem**

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Students asked me questions after class on during pauses (shyness)? I asked for feedback

➤ What happened in the past



➤ What is this class about?

- **General ideas** are relatively easy to grasp
- **Details** are complicated and require longer study
- Get the general idea, try to understand if it fits your problem



➤ Class materials

<https://albertotonda.github.io>

Alberto Tonda

BioNewsResearchProjectsPublicationsTeachingExperience

Teaching

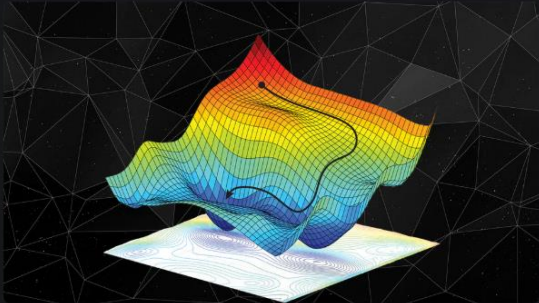


DEEP LEARNING

Deep learning in practice with pytorch

Class on deep learning for Ph.D. students enrolled in Université Paris-Saclay, taught in English

Apr 7, 2025



OPTIMIZATION

Optimization algorithms for artificial intelligence

Class on optimization for Ph.D. students enrolled in Université Paris-Saclay, taught in English

Apr 7, 2025

➤ Who am I?

- Career
 - Bachelor and Master in Computer Science Engineering
 - Ph.D. from Politecnico di Torino, Italy, in 2011
 - Permanent researcher in France since late 2012 (INRAE)
 - Senior researcher since 2023
- Research interests
 - Stochastic optimization, ML (xAI)
 - Applied to biological/agri-food data
 - Research: applied + algorithms



➤ What is optimization?

- Nearly every choice is an optimization problem
 - Shape of a car to minimize wind resistance
 - Values of the weights of a neural network to best perform a task
 - Weight distribution in a plane to minimize shaking
 - Pick a stock market portfolio to maximize revenue
 - Trace route inside a city to reach a point as fast as possible
 - Choose career that makes you satisfied and happy
 - ...



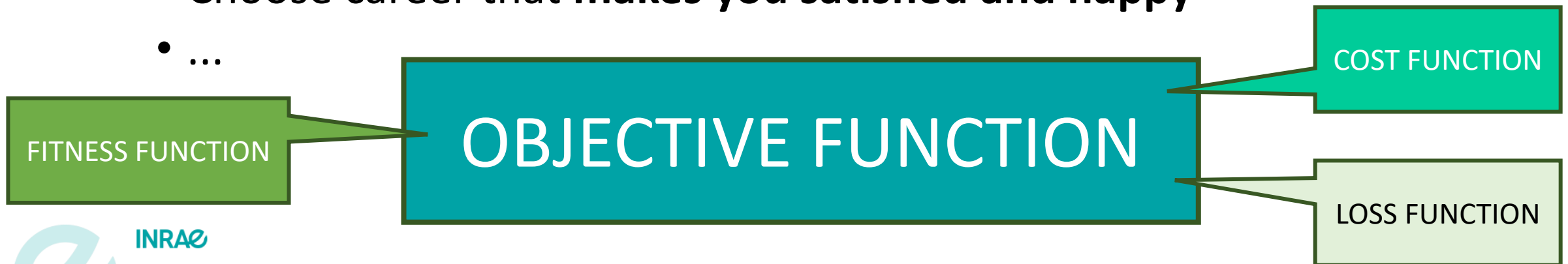
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OBJECTIVE FUNCTION

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➤ What is optimization?

- Objective function
 - Measure of goodness of a candidate solution
 - Quantitative, not qualitative (unless we can somehow sort it)
 - Good candidate solutions are usually close to other good solutions
 - If you pick the wrong objective function, you are screwed
- Candidate solutions
 - Possible inputs of the objective function
 - High-level representation that includes all possible solutions
 - Example: $\{x_0, x_1, \dots, x_n\}$ with $x_i \in [0,1] \cap \mathbb{R}$

➤ What is optimization?

- Given an objective function $y = f(x)$, find x^* such that

$$x^* = \underset{x}{\operatorname{argmax}}(y)$$

$$x^* = \underset{x}{\operatorname{argmin}}(y)$$

- Find the values of x that **maximize** or **minimize** y

➤ What is optimization?



➤ What is an algorithm?

➤ What is an algorithm?

- Series of steps to achieve an objective
 - Bake a cake
 - Sort the elements inside an array, by increasing value
 - Find the best possible input value for a given function
- Algorithmic complexity
 - Time to complete objective often depends on **size of input**
 - Baking ten cakes takes more time than baking one*
 - Sorting a large array takes more time than a small one

**unless you somehow perform everything in parallel with ten ovens*



➤ What is an algorithm?

- Algorithmic complexity is important for **expectations**
 - In particular, worst-case scenario (upper bound)
 - Notation: $O(f(n))$ where n is the size of the input
- Polynomial time vs Super-polynomial time
 - $O(n^k)$ vs $O(2^{n^k})$ or $O(n!)$, “easy” vs “difficult”
 - In practice, polynomial time can be used for large instances
 - Super-polynomial time can only be used for small input size

➤ What is an algorithm?

- Computer scientists often look at worst-case scenario
 - In some cases, average-case scenario might be better
 - E.g. worst-case scenario exponential time, average polynomial
- Most real-world problems are Nondeterministic Polynomial
 - Checking if a solution is correct performed in polynomial time
 - Finding the best solution takes super-polynomial time



➤ What is Artificial Intelligence?

➤ What is Artificial Intelligence?

- John McCarthy, one of the founding fathers of Artificial Intelligence
- «*I invented this term Artificial Intelligence [...] because [...] we were **trying to get money** for a summer study [...] in 1956 [...]*»



➤ What is Artificial Intelligence?

- Short answer, there is no clear definition
 - We do not have a good definition of *intelligence*, so...
 - Broadly speaking, AI defines a *field* more than a *method*
 - Machine learning, reinforcement learning, symbolic AI, ...
- Tentative definitions (there is no agreement)
 - «When a non-biological being successfully completes a task commonly believed to require biological intelligence»
 - «Perceiving, synthesizing, and inferring information»
- How do we *measure* intelligence?



➤ What is Artificial Intelligence?

NARROW / WEAK

Focused on a specific task

- Symbolic AI
 - E.g. rule-based systems
- Machine learning
 - Supervised, unsupervised
 - Natural language processing
 - Image recognition/segmentation
- Reinforcement learning
- Neuro-symbolic AI

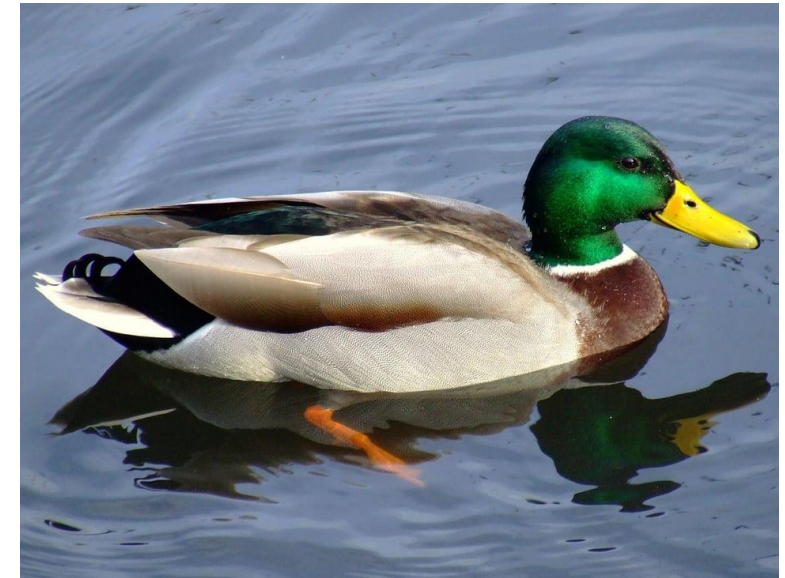
GENERAL (AGI)

Can perform any type of (human?) task

- Does not exist (...yet?)
- Closest thing is NLP: Large Language Models (LLM) like ChatGPT

➤ Symbolic AI

- Symbolic manipulation
 - Reality is *continuous* (with good approximation)
 - Symbols are *discrete*, and humans are good at using them



➤ Symbolic AI

- Symbols seem normal and natural, map into the real world (in linguistics, it's called *extension*)
- Natural language is a powerful human symbol manipulator
- However, there is chaos hidden under the surface
 - What is the reality of a *river*?
 - What is the reality of a *chair*?
 - What is the reality of a *number*?

➤ Symbolic AI

- Symbol can be hard to define, but we grasp it intuitively
 - It's an old, old problem: see Plato and Diogenes
 - *Entire fields of research* on this (neuroscience, cognitive sciences, neurolinguistics, ...)
- “Explaining” symbols to AI is harder yet
- Issues with “common sense”
- Reached limits in the 1980s

Man is but a featherless biped



Behold!
I've brought
you a man



➤ Symbolic AI

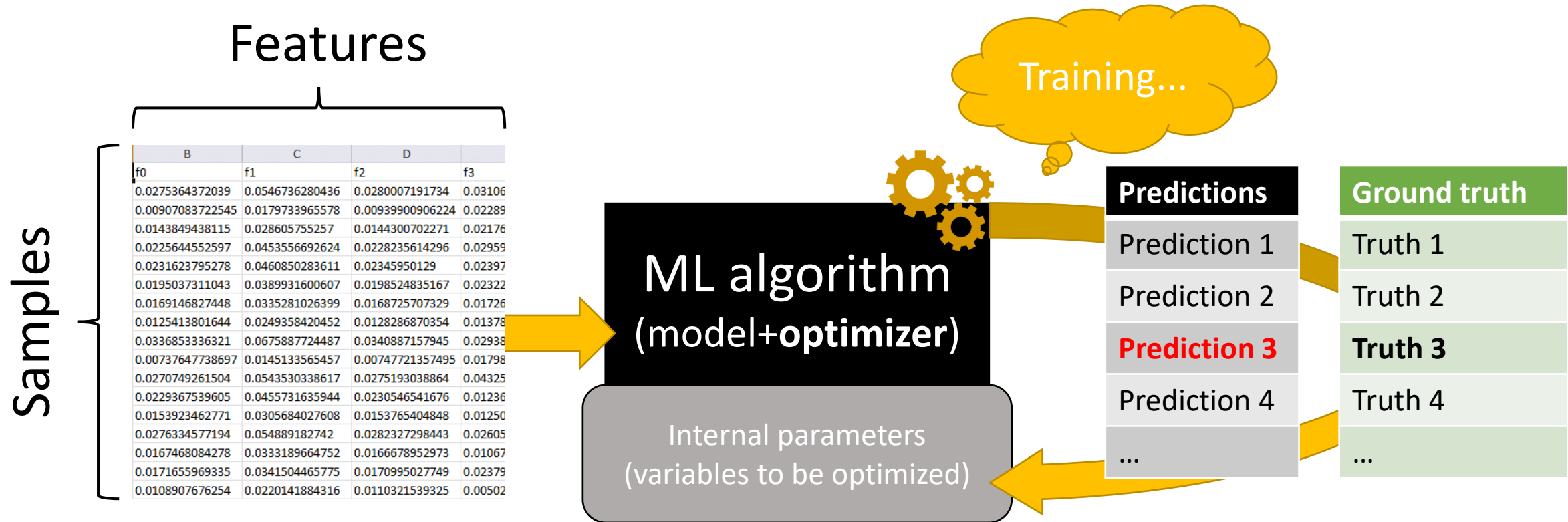
- In practice, find or exploit human-readable rules
 - Expert systems (“if-then-else” rules)
 - Knowledge graphs, linking entities with relationships
 - First-order logic rules
 - Decision trees (that are also considered part of ML!)
- Before the advent of ML, considerable success stories
- Symbolic AI is still in use, paired with ML

➤ Machine learning

- Learn a task directly from examples
 - No need for symbols, just large quantities of data
 - *Samples* (rows) and *features* (columns)
- “Dirty secret” of ML: it’s mostly optimization
 - Restate **learning task** as **optimization task**
 - Solve it relying on available (training) data



➤ Machine learning (supervised)



➤ Machine learning

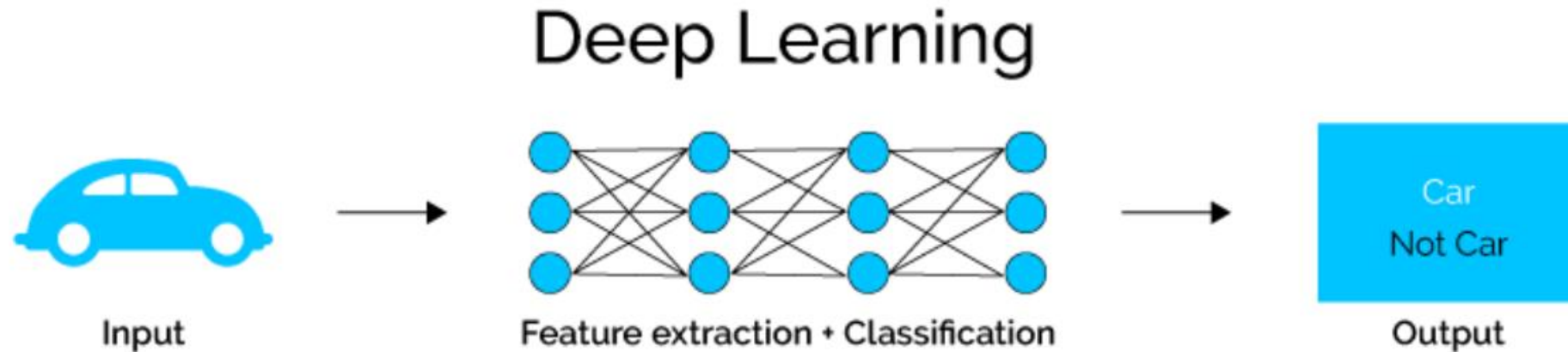
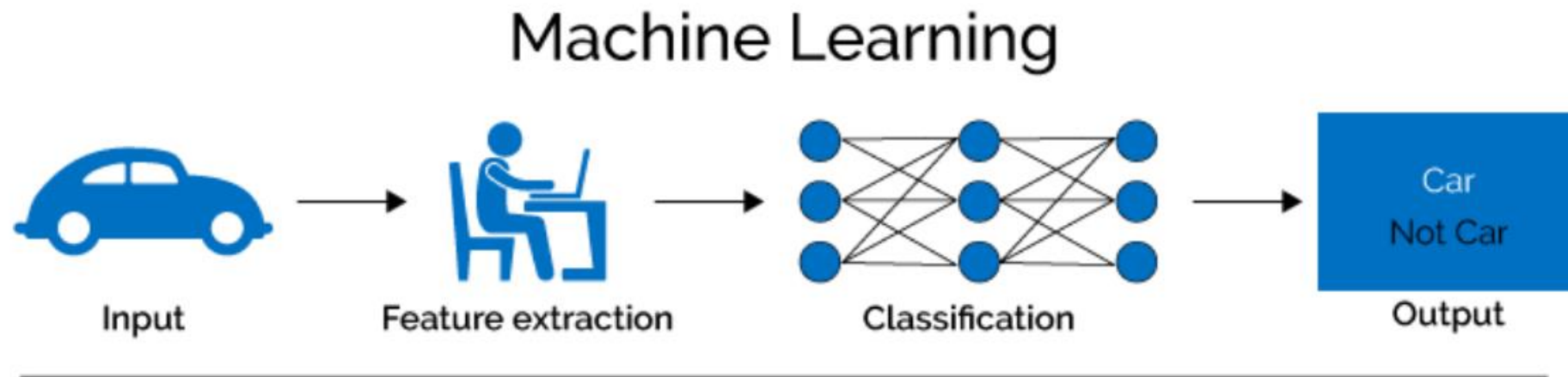
- Dominant paradigm since the 90s
 - First, artificial neural networks (ANNs or NNs)
 - Then, statistical learning algorithms
 - Decision trees (and ensembles of) and polynomial models
- Feature engineering
 - Works (still) well for **tabular data** (e.g. Excel spreadsheet)
 - Huge issues with **relational data** (e.g. images, text, sound...)
 - Hand-crafted features (not very successful)



➤ Machine learning

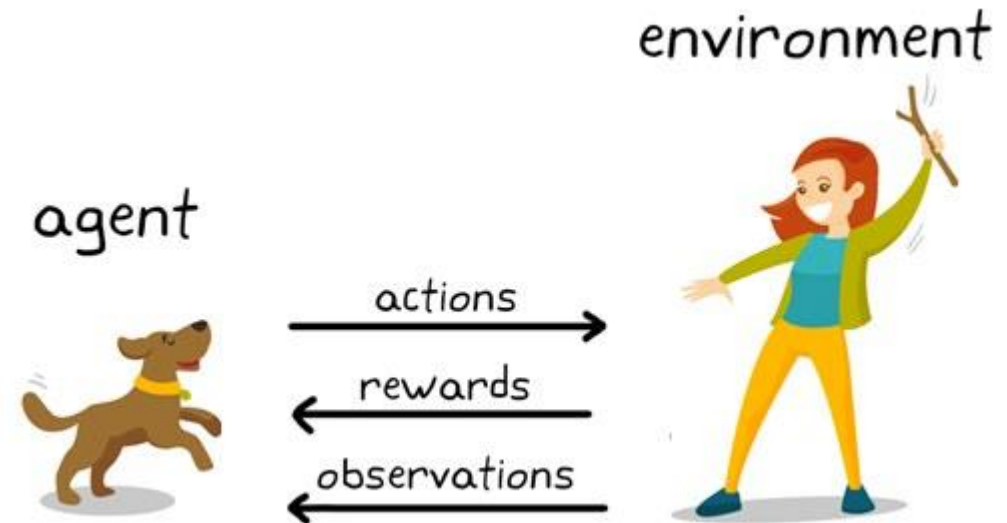
- Trade-off effectiveness/interpretability (black-box effect)
 - Good predictive models are extremely complicated
 - A single decision tree can be interpreted, 300 trees cannot
 - Same goes for polynomial models with 300 features
- Deep learning
 - General idea is that deep ANNs can automatically infer *features*
 - Convolutional NNs, Recurrent NNs, Transformers, ...
 - Fantastic success stories for **relational data**!

➤ Machine learning



➤ Reinforcement learning

- Similar to ML, but not exactly
 - No value for a *single* decision; reward after *series* of decisions
 - Learn a **policy** which tells you what to do from the **state** you are in
 - Example: chess game; is trading a Queen for a Knight good? Well, it depends on the board **state**



➤ Neuro-symbolic AI (NeSy)

- Might look complex, but the general idea is intuitive
 - Use neural/ML approach to map from data to symbols
 - Use symbolic AI to reason on symbols
 - (possibly) Go back to data using another neural/ML approach
 - Promise: **Effectiveness** (ML) + **Explainability** (Symbolic AI)
 - However, it's pretty hard to do, and problem-specific
- Interestingly, some of the biggest ML successes are NeSy
 - AlphaGo uses a mix of symbolic exploration and NN
 - AlphaFold is a mix of ~30 algorithms (some Symbolic, some ML)

➤ Artificial General Intelligence

- Hypothetical artificial intelligent agent
 - “Can learn (rapidly and cheaply) to perform any task that a human or another animal could perform, with minimal amounts of errors”
 - **It does not exist**, there is no clear path towards it
 - Lots of people scared by apparently quick advances of AI
 - Even some real experts (!!!)
- **My opinion**
 - Existential risk is non-existent
 - Real risks are already here, from misuse/misunderstanding



➤ Is optimization a kind of AI?

- Debatable, some experts would say “yes”
- My opinion: optimization is the **engine** of AI
- ...this is almost sure for ML



➤ Relationship between AI and optimization?

- Most “Intelligence” requires making good choices
 - Predict the next value in a time series, as precisely as possible
 - Correctly identify the human poses in a video
 - Make the best possible chess move, given the situation
 - Maximize your score in Super Mario
 - Generate a sequence of words that best follows the input
 - Create the painting that best corresponds to a written prompt

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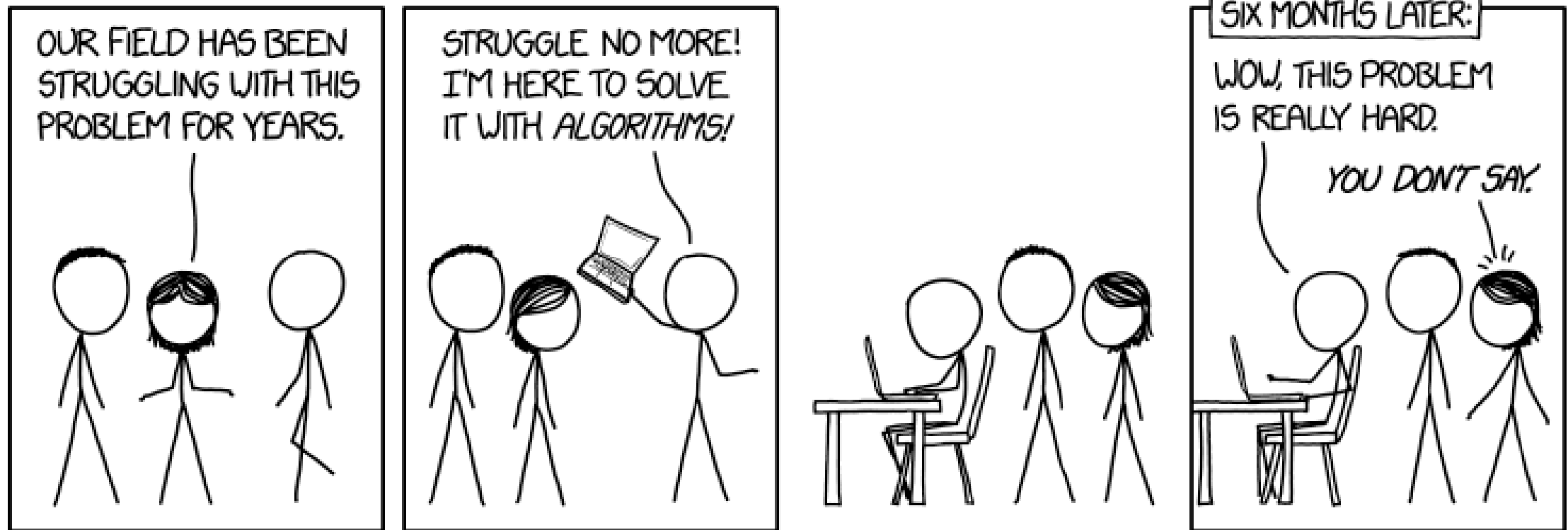
➤ Relationship between AI and optimization?

- AI: from many (many!!!) possible choices, pick the best
- If you can evaluate your choices, you can optimize
- (**AFAIK**) **All AI systems** include some type of optimization, with the possible exception of hand-crafted ones

➤ Is it always good to optimize?

- Optimizing one objective might lead to undesired outcomes
 - Supply chain optimized for efficiency is *fragile*
 - “Robust optimization” considers perturbations, but how big?
- Multi-objective optimization can be helpful
 - Multiple conflicting objectives, improve one $\leftrightarrow$ worsen others
 - Find **optimal** trade-offs between objectives

➤ Conclusions



“XKCD” by Randall Munroe www.xkcd.com

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➤ Questions?